
Lab Guide: Monitor EC2 Security Group Changes Using AWS Config + SNS Email Alerts

Objective:

Track any changes made to EC2 Security Groups (SG) using **AWS Config** and send **email notifications** via **Amazon SNS**.

Prerequisites:

- An AWS account with admin access
 - A working email address to receive alerts
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Step 1: Create an SNS Topic for Notifications

1. Go to **SNS Console** → <https://console.aws.amazon.com/sns>
2. Click **Topics** → **Create topic**
3. Choose:
 - **Type:** Standard
 - **Name:** `EC2SecurityGroupChangesTopic`
4. Click **Create topic**

Step 2: Subscribe Email to the Topic

1. In the topic details page, click **"Create subscription"**
2. Set:
 - **Protocol:** Email
 - **Endpoint:** Your email address (e.g., `your.email@example.com`)
3. Click **Create subscription**
4. Go to your email and **confirm the subscription**

Step 3: Create AWS Config Recorder

1. Go to **AWS Config Console** → <https://console.aws.amazon.com/config>
2. If this is your first time:
 - Click **"Get started"**
3. **Resource types to record:**
 - Choose **Specific resource types**
 - Select:
 - `EC2: SecurityGroup`
4. **Delivery method:**
 - AWS Config requires an **S3 bucket**.
 - You can create a new bucket: `config-logs-yourname`
5. **IAM Role:**

- Let AWS create a role automatically.
6. Click **Next** → **Confirm** to enable AWS Config
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Step 4: Set Up a Config Rule to Track SG Changes

1. In the **AWS Config Console** → Click **Rules** → **Add rule**
 2. Search for and select:
 - **changed-resource**
 3. Set:
 - **Trigger:** Configuration changes
 - **Resource type:** **AWS::EC2::SecurityGroup**
 4. In **Actions**, click **Add action**:
 - Choose **SNS topic**
 - Select **EC2SecurityGroupChangesTopic**
 5. Click **Next**, then **Add rule**
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Step 5: Test the Setup

1. Go to **EC2 Console** → **Security Groups**
2. Modify an inbound/outbound rule (e.g., add a new SSH rule)
3. Wait a few minutes
4. Check:

- AWS Config will log the change
- SNS will send an **email notification** to your address

Summary

Component	Purpose
AWS Config	Monitors and records changes to SGs
SNS Topic	Triggers email notifications
Email Sub	You receive alerts when changes occur

Cleanup (Optional)

To avoid extra charges:

- Delete the **SNS topic**
 - Stop **AWS Config recorder**
 - Delete **S3 bucket** (only if not reused)
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